

EECS 4080 – Supervisor Feedback

Milestone 2a: Data Lock *(Reviewed via email submission and GitHub repository)*

Overall Assessment

Milestone 2a is **partially complete**. Your core data pipeline is solid: you have collected datasets from appropriate sources, correctly merged and deduplicated them, and your frozen train/test splits have been committed to the GitHub repository with stratified class balance verified. These are the most critical requirements of Data Lock, and you have satisfied them. However, three gaps remain that you must address before proceeding to Milestone 2b baseline work: a missing validation split, an incomplete EDA (attack-type breakdown is absent), and an empty README. The missing validation split in particular will create data leakage risk if you add it later without re-freezing.

What Went Well

Strengths

- Good dataset selection: deepset/prompt-injections is the benchmark dataset named in your signed contract. Adding rogue-security/prompt-injections-benchmark to increase corpus size is a sensible and well-motivated choice.
- Your pipeline is clean: label mapping (string to integer), concatenation, and deduplication are all handled correctly, yielding a final corpus of 5,517 rows.
- Stratified splitting is the right approach for this imbalanced binary task. The ~60.3% / 39.7% distribution being consistent across both your train and test sets confirms stratification worked correctly.
- Your frozen CSVs are committed to the data/ folder in the public repo. This is exactly what the contract required — your splits are now the single source of truth and isolated from the raw data.
- Your EDA notebook (notebooks/eda.ipynb) is present and the word count distribution histogram is a useful structural insight: shorter prompts tend to be benign, longer prompts more likely malicious. **This is an interesting finding worth discussing in your final report.**
- Your repo is public, structured with logical folders (data/, notebooks/, src/, results/), and a requirements.txt is present.

Gaps and Required Actions

1. Missing Validation Split : Action Required Before Milestone 2b

Your contract specifies “train/validation/test dataset splits” (plural). You have only produced a train and test set. You will need a validation split for hyperparameter tuning during DistilBERT fine-tuning in Milestone 3 for setting learning rate, batch size, and early stopping criteria. If you carve out the validation set from your training data later (after running Milestone 2b baselines), it creates a sequencing problem and a potential leakage risk. Please freeze and commit it now, alongside your existing splits.

Required Action

- Re-split your training data to produce a three-way frozen split: train (~64%), validation (~16%), test (~20%) of the full corpus, all stratified.
- Commit three frozen CSVs: frozen_train_set.csv, frozen_val_set.csv, frozen_test_set.csv.
- Please do not proceed to your Milestone 2b baseline runs until this is done, as your baselines must be run on the frozen test set only.

2. EDA Incomplete — Attack-Type Breakdown Missing

Your contract requires EDA to map “attack-type breakdowns” not just binary class balance. Your current EDA documents how many prompts are good vs. bad, which satisfies class balance. But it does not categorise the types of injection attacks present in your dataset. Both of your benchmark datasets contain prompts representing different attack sub-types (e.g., instruction override, role-play manipulation, jailbreaking, goal hijacking). Understanding their distribution is important because it determines what your model is actually being trained to detect, and what gaps may exist in your evaluation.

Required Action

- Inspect both of your source datasets for any attack sub-type labels or tags (Hugging Face dataset cards often include this metadata).
- If sub-type labels are not present in the data, use keyword-based heuristic tagging to approximate attack categories (e.g., prompts containing 'ignore previous', 'pretend you are', 'DAN', etc.).
- Add a section to your EDA notebook that plots the distribution of attack sub-types across your training set.
- You will need this analysis before Milestone 5 (benchmarking) regardless — better to do it now while your data pipeline is fresh.

3. README is Empty

Your README.md file is 0 bytes. it is completely empty. This is the first thing any reader will see when visiting your repository. A thorough README is also listed as a final deliverable in your contract. While a complete README is not strictly a Milestone 2a requirement, starting with an empty one is something you should fix now.

Required Action

- Add a README.md with at minimum: project title and description, folder structure explanation, instructions for reproducing your data pipeline (how to run your src/ scripts and notebooks), dataset sources with links, and environment setup (referencing your requirements.txt).
- This can be brief at this stage but must not be empty.

4. Minor: .DS_Store Committed to Repository

You have committed a macOS system file (.DS_Store) to your repo. This is a minor issue but suggests your .gitignore was not configured before your initial commit. Please clean this up.

Required Action

- Run: git rm --cached .DS_Store, then commit the removal.

- Add `.DS_Store` to your `.gitignore` (if not already there) to prevent it from being committed again.

Before Proceeding to Milestone 2b

Please complete the following before running any baseline metrics on the frozen test set:

1. Create and commit a frozen validation split (`frozen_val_set.csv`) alongside your existing train and test CSVs.
2. Add the attack-type breakdown analysis to your notebooks/`eda.ipynb` and commit.
3. Add content to your `README.md` — at minimum, a project description, folder structure, and data pipeline instructions.
4. Remove `.DS_Store` from your repo (`git rm --cached`) and ensure it is in your `.gitignore`.